

Week 5 Exercise 1

Execution Trace Table

Mini CPU : Compute $(A + B) - C$

Cycle	State	PC	IR	ACC	ADDR	MEM_DATA_OUT	Action
1	FETCH1	0	–	00	00	stale	Begin fetch
2	FETCH2	0	10	00	00	10	IR $\leftarrow$ LOAD 0x10, PC=1
3	EXECUTE	1	10	00	10	stale	Read A
4	WRITEBACK	1	10	0F	10	0F	ACC $\leftarrow$ 15
5	FETCH1	1	10	0F	01	stale	Fetch ADD B
6	FETCH2	1	31	0F	01	31	IR $\leftarrow$ ADD 0x11
7	EXECUTE	2	31	0F	11	stale	Read B
8	WRITEBACK	2	31	18	11	09	ACC=15+9=24
9	FETCH1	2	31	18	02	stale	Fetch ADD(-C)
10	FETCH2	2	32	18	02	32	IR $\leftarrow$ ADD 0x12
11	EXECUTE	3	32	18	12	stale	Read -4
12	WRITEBACK	3	32	14	12	FC	ACC=24+(-4)=20
13	FETCH1	3	32	14	03	stale	Fetch STORE
14	FETCH2	3	53	14	03	53	IR $\leftarrow$ STORE
15	EXECUTE	4	53	14	13	stale	Prepare RAM write
16	WRITEBACK	4	53	14	13	00	RAM[result]=0x14
17	FETCH1	4	53	14	04	stale	Fetch HALT
18	FETCH2	4	60	14	04	60	IR $\leftarrow$ HALT
19	HALT	5	60	14	–	–	DONE=1